

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, r2_score
```

```
df = pd.read_csv("StudentsPerformance.csv")
```

```
print(df.head())
print(df.info())
print(df.columns)
```

```
   gender race/ethnicity parental level of education   lunch \
0  female      group B      bachelor's degree   standard
1  female      group C      some college   standard
2  female      group B      master's degree   standard
3   male      group A  associate's degree  free/reduced
4   male      group C      some college   standard

   test preparation course  math score  reading score  writing score
0                none           72           72           74
1            completed           69           90           88
2                none           90           95           93
3                none           47           57           44
4                none           76           78           75
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 8 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   gender                                1000 non-null  object
1   race/ethnicity                        1000 non-null  object
2   parental level of education           1000 non-null  object
3   lunch                                  1000 non-null  object
4   test preparation course               1000 non-null  object
5   math score                            1000 non-null  int64
6   reading score                         1000 non-null  int64
7   writing score                          1000 non-null  int64
dtypes: int64(3), object(5)
memory usage: 62.6+ KB
None
Index(['gender', 'race/ethnicity', 'parental level of education', 'lunch',
      'test preparation course', 'math score', 'reading score',
      'writing score'],
      dtype='object')
```

```
print(df.isnull().sum())
```

```
gender                0
race/ethnicity        0
parental level of education  0
lunch                 0
test preparation course  0
math score            0
reading score         0
writing score         0
dtype: int64
```

```
df = pd.get_dummies(df, drop_first=True)
```

```
y = df['math score']
```

```
X = df.drop('math score', axis=1)
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42
)
```

```
model = LinearRegression()

model.fit(X_train, y_train)
```

▼ LinearRegression ⓘ ?

```
LinearRegression()
```

```
y_pred = model.predict(X_test)
```

```
mae = mean_absolute_error(y_test, y_pred)

r2 = r2_score(y_test, y_pred)

print("Mean Absolute Error:", mae)
print("R2 Score:", r2)
```

```
Mean Absolute Error: 4.214763142474851
R2 Score: 0.8804332983749565
```

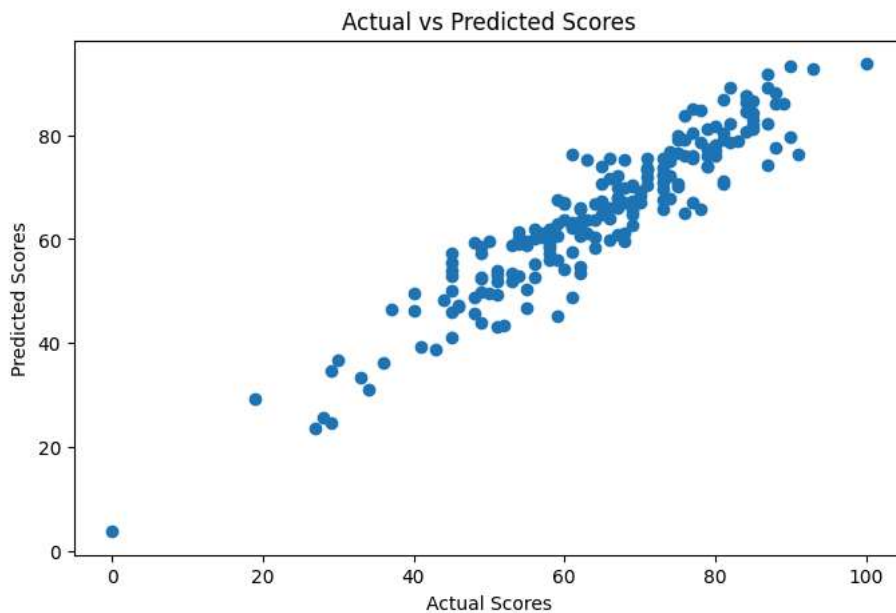
```
plt.figure(figsize=(8,5))

plt.scatter(y_test, y_pred)

plt.xlabel("Actual Scores")
plt.ylabel("Predicted Scores")

plt.title("Actual vs Predicted Scores")

plt.show()
```



1. Reading and writing scores strongly affect math scores.
2. Students completing test preparation perform better.
3. Linear Regression provided reasonable prediction accuracy.

CONCLUSION

The project successfully predicted student math scores using machine learning techniques. The analysis identified important factors influencing student performance.

